

量子力学期末考试 B 卷参考答案

第一题

(1) $N = \frac{1}{3}$, $E_n = -Ry\frac{1}{n^2}$, $n = 2$ 的几率为 $\frac{8}{9}$, $n = 3$ 的几率为 $\frac{1}{9}$

(2) 几率为 $\frac{2}{3}$

(3) $2\hbar^2$

(4) 波函数为 $\Psi(\vec{r},0) = \frac{1}{3} \left[\sqrt{2}\psi_{2,1,-1}(\vec{r}) - \psi_{2,1,0}(\vec{r}) + \sqrt{5}\psi_{2,1,1}(\vec{r}) \right] e^{-iE_2t/\hbar} + \frac{1}{3}\psi_{3,1,1}(\vec{r})e^{-iE_3t/\hbar}$

第二题

采用矩阵表示

$$\hat{H} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$|\alpha\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}} \quad |\beta\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}$$

$$\hat{O} = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \rightarrow |\psi(0)\rangle = |\beta\rangle$$

$$|\psi(0)\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}} \quad |\psi(t)\rangle = \frac{e^{-it}|+\rangle - e^{it}|-\rangle}{\sqrt{2}}$$

$$|\langle\alpha | \psi(t)\rangle|^2 = \left| \left(\frac{\langle+| + \langle-|}{\sqrt{2}} \right) \left(\frac{e^{-it}|+\rangle - e^{it}|-\rangle}{\sqrt{2}} \right) \right|^2 = \sin^2 t \quad t = (2n-1)\frac{\pi}{2}$$

第三题

S_z 的本征态为 $|\pm\rangle$

$$|\psi\rangle = a|+\rangle + b|-\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$a > 0 \quad b = |b|e^{i\theta}$$

$$\frac{2}{\hbar}\langle S_x \rangle = 2a\sqrt{1-a^2}\cos\vartheta$$

$$\frac{2}{\hbar}\langle S_y \rangle = 2a\sqrt{1-a^2}\sin\vartheta$$

当 $\alpha^2 + \beta^2 < 1$ 时, 存在两个相对于 $a^2 = \frac{1}{2}$ 对称的解,

当 $\alpha^2 + \beta^2 = 1$ 时, $a = \frac{1}{\sqrt{2}} = |b|$, $|\psi\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}e^{i\vartheta}|-\rangle$ 几率分别为 $\frac{1}{2}$

$$\langle S_x \rangle = \frac{\hbar}{2}(a^*b^*) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \frac{\hbar}{2}(a^*b + ab^*)$$

$$\langle S_y \rangle = \frac{\hbar}{2}(a^*b^*) \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = -\frac{\hbar}{2}i(a^*b - ab^*)$$

$$|a|^2 + |b|^2 = 1 \Rightarrow |b|^2 = 1 - a^2 \Rightarrow b = \sqrt{1 - a^2}e^{i\theta}$$

$$\alpha^2 + \beta^2 = \frac{4}{\hbar^2} \left(\langle S_x \rangle^2 + \langle S_y \rangle^2 \right) = 4a^2(1 - a^2), \quad a^2 \leq 1$$

$$\alpha^2 + \beta^2 \leq 4a^2(1 - a^2) \Big|_{a^2=\frac{1}{2}} = 1$$

第四题

(1) 基态能量和波函数

$$E_1^0 = \frac{\pi^2 \hbar^2}{32ma^2}$$
$$\psi_1^0 = \frac{1}{\sqrt{2a}} \cos \frac{\pi x}{4a}$$

(2) 一级微扰能量

$$\begin{aligned} E^{(1)} &= \frac{V_0}{2a} \int_{-a}^a \cos^2 \frac{\pi x}{4a} dx = \frac{V_0}{2a} \int_{-a}^a \frac{1}{2} \left(1 + \cos \frac{\pi x}{2a} \right) dx \\ &= \frac{V_0}{4a} (x)_{-a}^a + \frac{V_0}{4a} \frac{2a}{\pi} \left(\sin \frac{\pi x}{2a} \right)_{-a}^a \\ &= \frac{V_0}{2} + \frac{V_0}{\pi} = V_0 \left(\frac{1}{2} + \frac{1}{\pi} \right) \end{aligned}$$

$$E_1 = \frac{\pi^2 \hbar^2}{32ma^2} + V_0 \left(\frac{1}{2} + \frac{1}{\pi} \right)$$

第五题

电子处于自旋三重态 $|S = 1, M_s\rangle$, 空间波函数必须是反称的。取质心坐标系,

$$\begin{aligned} X &= \frac{1}{2}(x_1 + x_2), & P &= p_1 + p_2 \\ x &= x_1 - x_2, & p &= \frac{1}{2}(p_1 - p_2) \end{aligned} \qquad H = \frac{P^2}{4m} + \frac{p^2}{m} + V(x)$$

$$\Psi(x_1, x_2) = \frac{1}{\sqrt{2\pi}} e^{iKX} \psi_\epsilon(x) \qquad \psi''_\epsilon(x) = \frac{m}{\hbar^2} [V(x) - \epsilon] \psi(x)$$

$$E = \frac{\hbar^2 K^2}{4m} + \epsilon$$

$$\Psi(x_1, x_2) = -\Psi(x_2, x_1) \implies \psi_\epsilon(-x) = -\psi_\epsilon(x)$$

$$\begin{aligned} \psi_{k,l}(x,y) &= \psi_k(x)\psi_l(y) \\ E_{k,l} &= \frac{\pi^2 \hbar^2}{8m} \left(\frac{k^2}{a^2} + \frac{l^2}{b^2} \right) \end{aligned} \qquad \psi_k(x) = \begin{cases} \sqrt{\frac{1}{a}} \cos \frac{k\pi x}{2a}, & \text{odd } k \\ \sqrt{\frac{1}{a}} \sin \frac{k\pi x}{2a}, & \text{even } k \end{cases} \qquad \begin{aligned} \psi_{k,l}(x,y) &= \sqrt{\frac{1}{ab}} \cos \frac{k\pi x}{2a} \sin \frac{l\pi y}{2b} \\ \psi_{k,l}(x,y) &= \sqrt{\frac{1}{ab}} \sin \frac{k\pi x}{2a} \cos \frac{l\pi y}{2b} \end{aligned}$$